

pipeline_report

EP

2026-08-04

Overview

Table 1: Summary of diversity and recording effort by city

city	n_cells	mean_richness	mean_shannon	mean_simpson	total_records	total_survey_days
Bristol	143	218.08	3.39	0.89	836079	84948
Exeter	58	121.90	3.41	0.92	218942	21875
Gateshead	80	63.86	2.86	0.83	223262	19009
Newcastle	109	105.07	3.15	0.88	318004	35752
Nottingham	99	100.35	3.22	0.89	199674	27043

Grid Validation

Grid geometry

Table 2: Grid geometry check: validity and cell area by city

city	n_cells	n_invalid_geom	mean_area	min_area	max_area
Bristol	143	0	100	100	100
Exeter	58	0	100	100	100
Gateshead	80	0	100	100	100
Newcastle	109	0	100	100	100
Nottingham	99	0	100	100	100

CRS used across all cities: EPSG:27700

Check for duplicate and missing cells

No duplicate cell IDs found within any city.

Check for missing values

Table 3: Count of missing (NA) values per predictor, by city

city	sp_richness	simpson	hannan	n_days_record	dtm_elev	avg_imperv	tree_hgt	weighted_precip	weighted_temp	dist_to_river	pct_green	grass	water	risk
Bristol	0	0	0	0	0	0	0	0	0	0	0	0	0	34
Exeter	0	0	0	0	0	4	0	0	0	0	0	0	1	13
Gateshead	0	0	0	0	0	0	0	0	0	0	0	0	12	32
Newcastle	0	0	0	0	0	0	0	0	0	0	0	0	2	46
Nottingham	0	0	0	0	0	0	0	0	0	0	0	0	1	37

Check that values are within a reasonable range

i.e. a percentage is between 0-100.

Table 4: Min/max values for numerical variables

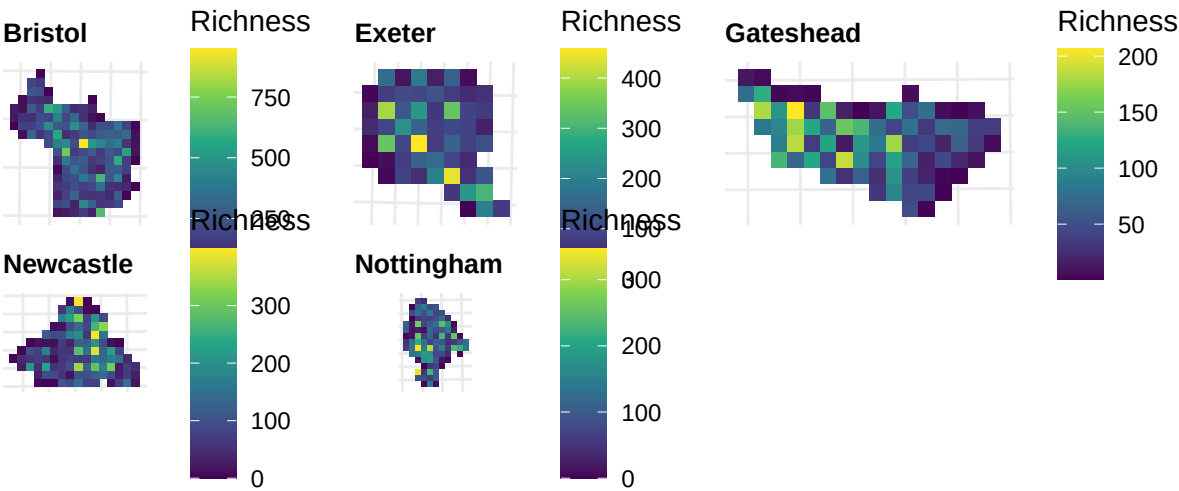
stat	value
sp_richness_min	0.00
sp_richness_max	946.00
simpson_min	0.00
simpson_max	1.00
n_days_min	1.00
n_days_max	3955.00
dtm_elev_min	1.06
dtm_elev_max	151.74
avg_imperv_min	0.43
avg_imperv_max	84.32
weighted_precip_min	639.13
weighted_precip_max	1016.80
weighted_temp_min	9.19
weighted_temp_max	11.10
dist_to_river_min	1.18
dist_to_river_max	2993.63
pct_green_min	0.00
pct_green_max	1.27

Cross-city consistency

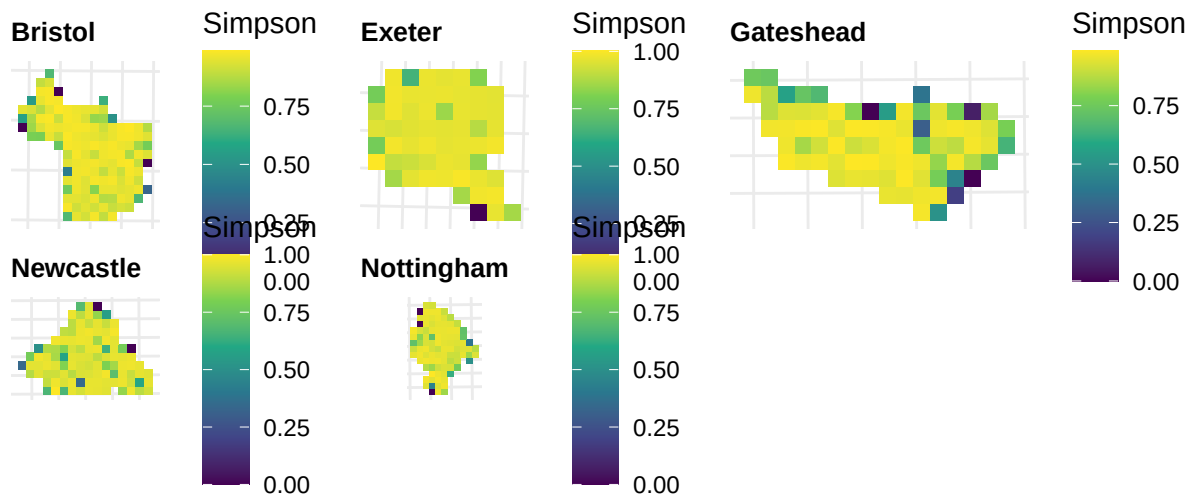
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## All cities have an identical column structure: TRUE
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Spatial patterns

Species richness



Simpson diversity



Flood Risk summary

Table 5: Cell counts by surface water flood risk (RoFSW) category and city

rofsw_risk	Bristol	Exeter	Gateshead	Newcastle	Nottingham
Low	94	51	68	93	94
Medium	19	0	0	0	1
High	30	6	0	14	3
NA	0	1	12	2	1

Table 6: Cell counts by river and sea flood risk (RoFRS) category and city

rofrs_risk	Bristol	Exeter	Gateshead	Newcastle	Nottingham
Very low	17	0	1	1	3
Low	24	14	11	13	28
Medium	3	2	4	1	4
High	65	29	32	48	27

rofrs_risk	Bristol	Exeter	Gateshead	Newcastle	Nottingham
NA	34	13	32	46	37

Models

Species richness (negative binomial GLM)

Table 7: Full richness model coefficients

term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.171	1.022	0.167	0.867
n_days	0.001	0.000	10.736	0.000
dtm_elev	0.000	0.002	0.133	0.894
avg_imperv	0.005	0.003	1.582	0.114
rofrsw_riskMedium	0.108	0.259	0.419	0.675
rofrsw_riskHigh	0.163	0.139	1.171	0.242
rofrs_riskLow	-0.149	0.257	-0.579	0.563
rofrs_riskMedium	-0.056	0.330	-0.170	0.865
rofrs_riskHigh	-0.209	0.250	-0.838	0.402
avg_area_ha	0.003	0.004	0.778	0.437
treeht	-0.027	0.040	-0.670	0.503
weighted_precip	0.000	0.001	0.452	0.651
weighted_temp	0.374	0.137	2.723	0.006
dist_to_river	0.000	0.000	-0.618	0.536
pct_greenspace	1.734	0.360	4.816	0.000

Table 8: Richness: Null vs. Full models

logLik_null	logLik_full	AIC_null	AIC_full	chisq	df	p_value
-2879.981	-1788.753	5763.963	3609.505	2182.458	14	0

Simpson diversity (beta regression)

Table 9: Full Simpson model coefficients

term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.988	1.328	0.744	0.457
n_days	0.000	0.000	1.774	0.076
dtm_elev	-0.001	0.003	-0.196	0.845
avg_imperv	0.004	0.004	0.942	0.346
rofrsw_riskMedium	0.309	0.337	0.915	0.360
rofrsw_riskHigh	0.119	0.181	0.655	0.512
rofrs_riskLow	0.395	0.333	1.185	0.236
rofrs_riskMedium	0.494	0.430	1.149	0.251
rofrs_riskHigh	0.288	0.324	0.891	0.373
avg_area_ha	0.003	0.005	0.564	0.573

term	Estimate	Std. Error	z value	Pr(> z)
treeht	0.030	0.052	0.567	0.571
weighted_precip	0.000	0.001	-0.062	0.950
weighted_temp	-0.020	0.178	-0.115	0.909
dist_to_river	0.000	0.000	-0.685	0.494
pct_greenpace	1.019	0.471	2.163	0.031

Table 10: Simpson: Null vs. Full Models

logLik_null	logLik_full	AIC_null	AIC_full	chisq	df	p_value
396.25	277.201	-788.499	-522.402	-238.097	14	1

Recording effort

